

Investigations in Cardiology

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Investigations in Cardiology

Onuoha – Paediatrics, topic 8, Introduction to Cardiology / Investigations of the Cardiac System

Introduction

Investigations in cardiology are **tests or procedures for evaluation of cardiovascular diseases**.

They are carried out either at:

- **Bedside**, and/or
- **Laboratories**

Objectives of investigations in cardiology

1. Establish the diagnosis
2. Determine the aetiology
3. Identify the risk factors
4. Detect the complications
5. Monitor the disease progression
6. Evaluate associated co-morbidities
7. Make prognosis

List of investigations in cardiology

1. Chest X-Ray
2. Electrocardiogram
3. Electrophysiological Studies
4. Echocardiography (Cardiac Ultrasonography)
5. Cardiac Magnetic Resonance Imaging (CMRI)
6. Cardiac Catheterization
7. Coronary arteriography
8. Cardiac Enzymes and Troponins
9. Brain Natriuretic Peptide (BNP)
10. Lipid Profiles
11. Blood Sugar
12. Electrolyte, Urea and Creatinine (E&U)

Non-invasive versus invasive investigations

Non-invasive investigations – procedures or tests **limited to the body surface**, e.g. standard 12-lead ECG, transthoracic echocardiography.

Invasive investigations – procedures or tests that involve **penetration of the vascular system or body tissues**, e.g. electrophysiological studies, cardiac catheterization, transoesophageal echocardiography, angiography.

Chest X-Ray

Utilizes **ionizing radiation called X-Ray** for cardiac evaluation.

It is mainly useful in the study of **cardiac structure** and **changes in the pulmonary circulation which may be of cardiac origin**.

Uses of chest X-ray in cardiology

- To diagnose **cardiomegaly**
- To identify features of heart diseases such as **hypertension, heart failure, cardiomyopathy, congenital heart diseases, pericardial effusion, coarctation of aorta, valvular and pericardial calcifications**
- To diagnose **acute pulmonary oedema**
- To identify **precipitants of heart failure**, e.g. chest infection



Figure 1: Chest X-ray (PA view) showing the cardiac silhouette

Chest X-ray criteria for diagnosis of cardiomegaly in adults

1. **Cardiothoracic ratio (CTR) > 50%**
2. **Maximum transverse cardiac diameter > 15.5 cm**

Cardiothoracic ratio refers to the ratio of the **maximum transverse cardiac diameter** to the **maximum internal diameter of the thoracic cage**, expressed as a percentage.

Electrocardiogram (ECG)

The electrocardiogram is the **graphical record of the electrical activities of the heart obtained at the body surface**.

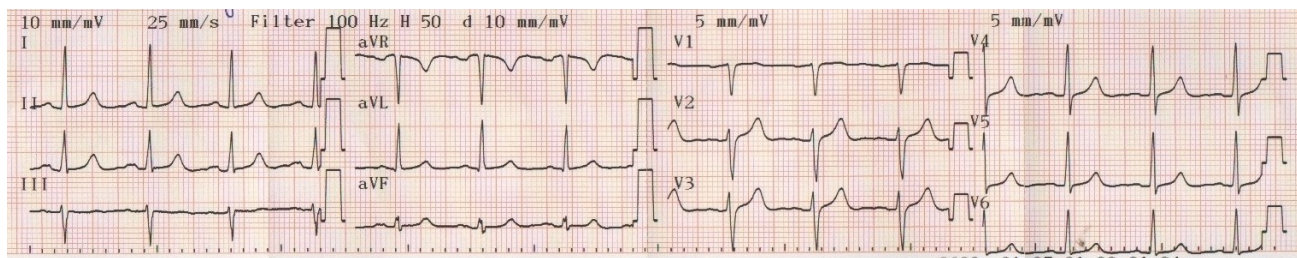


Figure 2: A standard 12-lead ECG

Forms of ECG

Standard 12-lead ECG

1. Utilizes **12 leads**
2. Recorded at rest, over a period of **10-20 seconds**
3. **Most widely used** form of ECG

Holter monitoring / ambulatory ECG – records ECG for **24 hours**

Stress ECG – ECG recorded **during exercise**

Indications for ECG

1. **Chamber abnormalities** e.g. ventricular hypertrophy, left atrial abnormalities
2. **Cardiac arrhythmias**
3. **Heart blocks** – atrioventricular blocks, bundle branch blocks
4. **Pre-excitation syndromes** e.g. Wolff-Parkinson-White syndrome
5. **Ischaemic heart diseases**
6. **Myocarditis**
7. **Cardiomyopathies**
8. **Pericardial diseases**
9. **Congenital heart diseases**
10. **Cardiac arrest**
11. **Electrolyte derangement** e.g. hyperkalaemia, hypokalaemia
12. **Drug effect** e.g. digoxin toxicity
13. **Pulmonary embolism**
14. **Cor pulmonale**



Figure 3: An ECG machine

Electrophysiological studies

Invasive procedures for studying **cardiac arrhythmias** and **pre-excitation syndromes** such as Wolff-Parkinson-White syndrome.

Echocardiography

Echocardiography is **cardiac imaging using ultrasound**. It is otherwise known as **cardiac ultrasound**.



Figure 4: An echocardiography machine

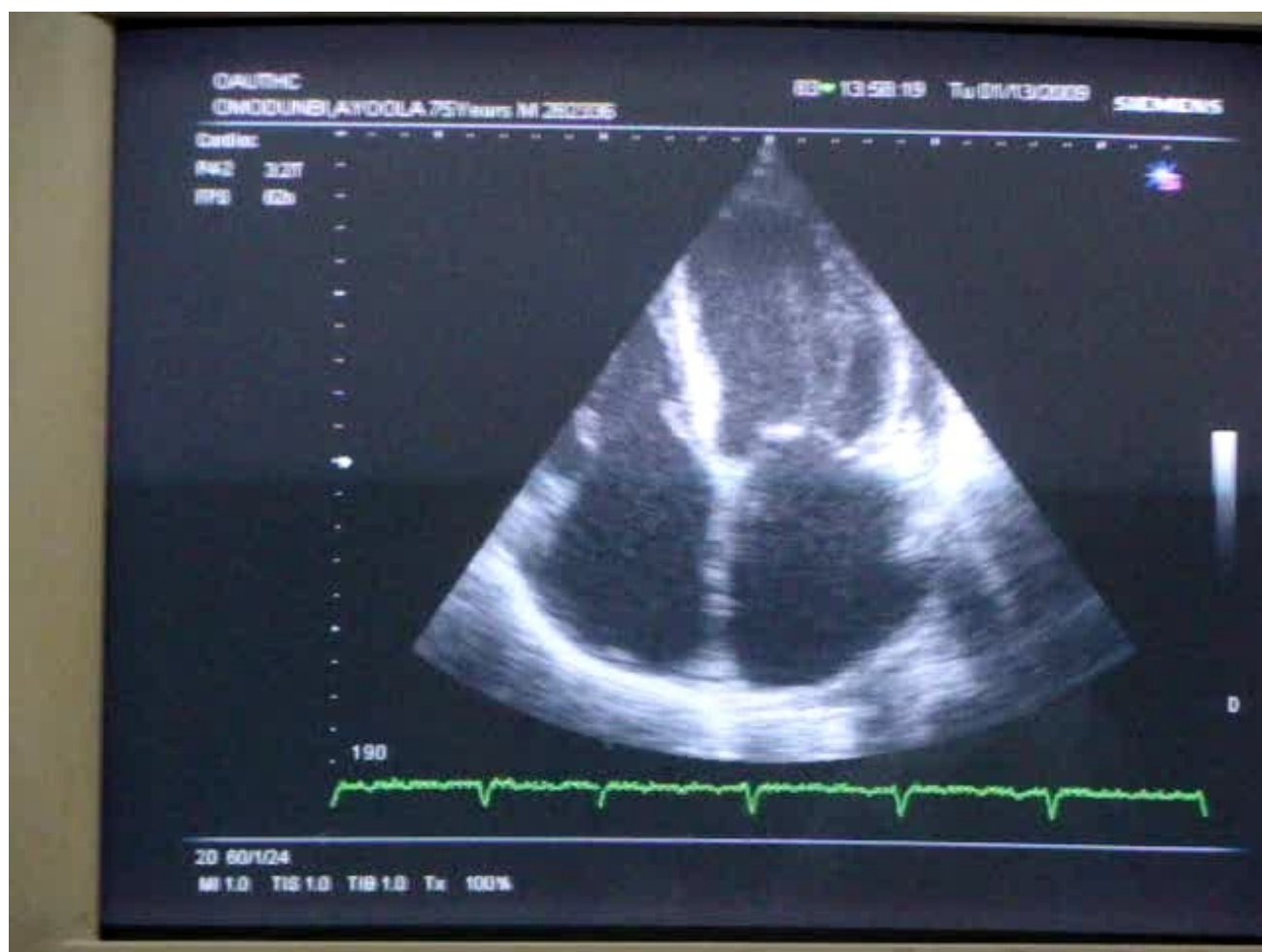


Figure 5: Echocardiography: apical 4-chamber view

Forms of echocardiography

Transthoracic echocardiography – utilizes a **transthoracic probe or transducer** to obtain cardiac imaging **from the body surface**

Transoesophageal echocardiography – utilizes a **transoesophageal probe or transducer** to obtain cardiac images **very close to the heart**

Uses of echocardiogram

- To evaluate **cardiac structures, functions and dysfunctions**
- To establish **diagnosis** of some cardiovascular diseases
- To detect **complications** of cardiovascular diseases

Indications for echocardiogram

Echocardiogram is useful in the evaluation of cardiovascular disorders such as:

1. Hypertensive heart diseases
2. Congenital heart diseases
3. Rheumatic heart diseases
4. Valvular heart diseases
5. Infective endocarditis
6. Cardiomyopathies
7. Pericardial diseases, especially **pericardial effusion**
8. Ischaemic heart disease
9. Heart failure
10. **Cardiac masses** e.g. intramural thrombus, atrial myxoma

Cardiac MRI

- Utilizes a **powerful magnetic field** to align the nuclear magnetization of **hydrogen atoms in water** in the body
- **Does not use ionizing radiation**
- Useful in the study of **cardiac structure**
- **Current gold standard for evaluation of left ventricular hypertrophy**

Cardiac catheterization

An **invasive** procedure involving **introduction of a catheter through the veins into the heart under fluoroscopy guidance**.

- Useful for evaluation of **intracardiac pressures**
- May also serve **therapeutic** value in the management of **valvular lesions or congenital heart diseases** such as atrial or ventricular septal defects

Coronary arteriography

An **invasive** procedure which involves the use of **contrast agents** for evaluation of the **coronary arteries**.

Useful in the **diagnosis and management of coronary artery diseases**.

Cardiac enzymes and troponins

Cardiac enzymes and troponins are **markers of myocardial injury**. The markers are **elevated in myocardial infarction**.

Cardiac enzymes include:

1. **Creatine phosphokinase (CK)**
2. **Aspartate aminotransferase (AST)**
3. **Lactate dehydrogenase (LDH)**

Troponins include: **troponin I** and **troponin T**

Brain Natriuretic Peptide (BNP) and NT-proBNP

- **BNP** is a **32-amino acid polypeptide** produced by the ventricles
- **NT-proBNP** is a **76 amino acid N-terminal fragment polypeptide** co-secreted with **BNP**
- Both are useful in the **screening and diagnosis of heart failure**
- They are very useful in **establishing prognosis of heart failure**

Other investigations

Other investigations are **supportive**, and are useful in detecting **risk factors, complications or co-morbidities**.

A. Fasting lipid profiles

Components include:

1. **Total cholesterol (TC)**
2. **Low density lipoprotein (LDL)**
3. **High density lipoprotein (HDL)**
4. **Triglycerides (TG)**

B. Blood sugar

To exclude the presence or not of **diabetes mellitus**.

Includes **fasting blood sugar (FBS)** and **2-hour postprandial sugar (2HPPS)**.

C. Electrolytes, urea and creatinine

Electrolytes include **Na⁺, K⁺, Ca²⁺, HCO₃⁻**.

Urea and creatinine are analyzed to:

1. Exclude effects of electrolytes as being responsible for the cardiovascular dysfunction
2. Detect whether or not **renal disease** is present in the patient